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MID SEMESTER EXAMINATION MARCH - 2025

B.Tech. (CSE)
Paper Name: **Pattern Recognition**

8th Semester
Paper Code: TCS 852

Time: 90 Mins

Maximum Marks: 50

Note:

- (i) Answer **all the questions** by choosing **any one of the sub questions**.
- (ii) Each questions carries 10 marks.

Q1. (2X10=20 Marks) (CO1)

- a) What is Machine perceptron? Explain with the help of an illustrative example.
- b) Imagine you are assigned to create a system for recognizing handwritten digits. Outline the key steps in the pattern recognition process, beginning with data acquisition and concluding with the generation of predictions.

Q2. (2X10=20 Marks) (CO2)

- a) What are discriminant functions, and how are they used in pattern recognition?
- b) Discuss the concept of the Normal density function. Describe the key characteristics of Normal distribution and how they affect the shape of the density curve. Provide examples to illustrate your explanation.

Q3. (2X10=20 Marks) (CO1 & CO2)

- a) What do you understand by Maximum Likelihood Estimation (MLE) explain with the help of example.
- b) Explain the core concepts of Bayesian decision theory and its application in decision-making under uncertainty.

Q4. (2X10=20 Marks) (CO2 & CO3)

- a) Explain the concept of unsupervised learning and its application in clustering.
- b) What does K-means clustering? Mention its key steps and how it works to partition a dataset into clusters.

Q5. (2X10=20 Marks) (CO2 & CO3)

- a) What distinguishes linear component analysis from non-linear component analysis?
- b) Explain the step-by-step process of Principal Component Analysis (PCA) and support your explanation with illustrative diagrams.

